

MASTER PROJECT IMPROVEMENTS & ARCHITECTURAL ROADMAP CHECKLIST

An Intelligent AI-Powered Restaurant Ordering System Using Augmented Reality

TCET Department of Computer Engineering | Capstone Evaluation (A.Y. 2026-27) | Team: Manav Singh, Sanskar Suryavanshi, Kesar Singh | Guide: Prof. Vinita Sunish

Production URL: <https://smart-restaurant-2za8.vercel.app/> | Executive 1-Page Comprehensive Project Analysis

1. SPATIAL AR/VR COMPUTING & 3D RENDERING • WebRTC Synchronized Multi-User AR: Peer-to-peer data mesh synchronizing 3D dish transform matrices at 60Hz so entire banquet parties view/rotate shared platters simultaneously. • NeRF & 3D Gaussian Splatting Pipeline: Instant 3D food asset reconstruction from a 45s chef smartphone video sweep, eliminating manual 3D modeling bottlenecks in <12 minutes. • WebGPU Acceleration & Sub-Millimeter Depth: Upgrading from WebGL to WebGPU for 120 FPS high-fidelity fluid/sauce rendering and micro-texture photorealism. • WebXR Ambient Light Estimation: Using `XRLightEstimate` to match real-world candlelit restaurant illuminance, shadows, and color temperature dynamically.	4. ENTERPRISE POS, ERP & BILLING INTEGRATION • Bidirectional POS Integration: Certified REST webhooks for Oracle Micros Symphony, Toast POS, Petpooja, and POSist for automated table order-to-terminal synchronization. • Automated Real-Time 86'ing: Instantaneous inventory depletion webhooks graying out sold-out items in diner AR menus to prevent awkward order cancellations. • Integrated Fiscal Billing & Cloud Checkout: Direct digital guest checks with automated GST/VAT calculation, split-bill math, and Apple Pay / Google Pay / UPI checkout. • Automated Cellar Vintage Depletion: Live cellar stock synchronization removing rare vintage bottles from sommelier recommendations upon uncorking.
2. ARTIFICIAL INTELLIGENCE & VISION QA • On-Device Generative AI Voice Sommelier: Lightweight edge LLM answering conversational pairing questions ("Which wine under ₹4k matches this ribeye?") with live cellar queries. • Computer Vision Kitchen Plating QA: Overhead 4K camera running YOLOv8/viT at the pass, checking volumetric mass, garnish placement, and sauce drizzle against the 3D standard. • Dynamic Multi-Allergen Safety Engine: Automated cross-contamination alerts and intelligent ingredient substitution recommendations based on guest profile history. • AI Dynamic Pricing & Menu Engineering: Real-time surge pricing algorithms optimizing dessert and specialty cocktail margins during peak dining hours.	5. IOT HARDWARE & SUSTAINABILITY ENGINEERING • IoT Tabletop BLE Load-Cell Weight Coasters: Continuous beverage level sensing beneath glassware dispatching silent refill prompts to waitstaff at 15% threshold. • 100% Zero-Paper Circularity: Complete elimination of laminated paper menus, saving kilograms of single-use plastic and reducing restaurant plate return waste by up to 28%. • Dynamic Low-Power Mobile Edge Delivery: Edge CDN caching and Draco mesh quantization saving device battery life and mobile network bandwidth. • Smart Table Occupancy & Turn Sensors: Low-power infrared sensors tracking table cleanliness and turnover readiness for floor management.
3. KITCHEN DISPLAY SYSTEM (KDS) & STATE ENGINE • Enterprise WebSocket / Redis Cluster: Upgrading local state synchronization to a distributed multi-terminal cloud pub/sub cluster for zero-latency multi-kitchen routing. • Predictive Prep-Time & Station Load Balancing: AI algorithms predicting dish preparation duration and balancing ticket routing across grill, pantry, and dessert lines. • Wearable Staff Haptic Notifications: Pushing silent smartwatch vibrations and table call alerts to waitstaff when dishes pass visual QA and are ready for runner dispatch. • Acoustic Station-Specific Chime Differentiation: Synthesizing unique frequency chimes for rush orders, allergen warnings, and beverage bar dispatches.	6. STRATEGIC PIVOT TO LUXURY FINE DINING • Strategic Pivot from Midrange to Fine Dining: Shifting focus from fast-turn casual dining (AOV ₹500) to luxury bistros (AOV ₹1,800-₹5,500+) where 3D depth drives high-margin spend. • Reset Survey Execution: Executing redesigned research survey across N=150 luxury gastronomy diners and N=25 Michelin/fine-dining executive chefs. • Multi-Course Chef Tasting Menu Journey: Curating sequential 5-to-7 course AR walkthroughs that guide diners through amuse-bouche, entrees, and dessert wine flights. • Luxury Ambiance White-Tablecloth UI: Refined minimalist typography, dark luxury mode, and discreet guest interaction respecting formal dining etiquette.
Phase 1 (Live Now): Zero-Install WebXR • 1:1 Metric Clamping • Live KDS • Acoustic Chimes • Vercel CDN	Phase 2 (Horizon 1): Luxury Dining Pivot • AI Voice Sommelier • Survey Reset Execution
Phase 3 (Horizon 2): WebRTC Multi-User Table AR • Automated NeRF / Gaussian Splatting 3D Pipeline	Phase 4 (Horizon 3): Kitchen Plating Vision QA Gate • Oracle Micros / Toast POS ERP Connectors